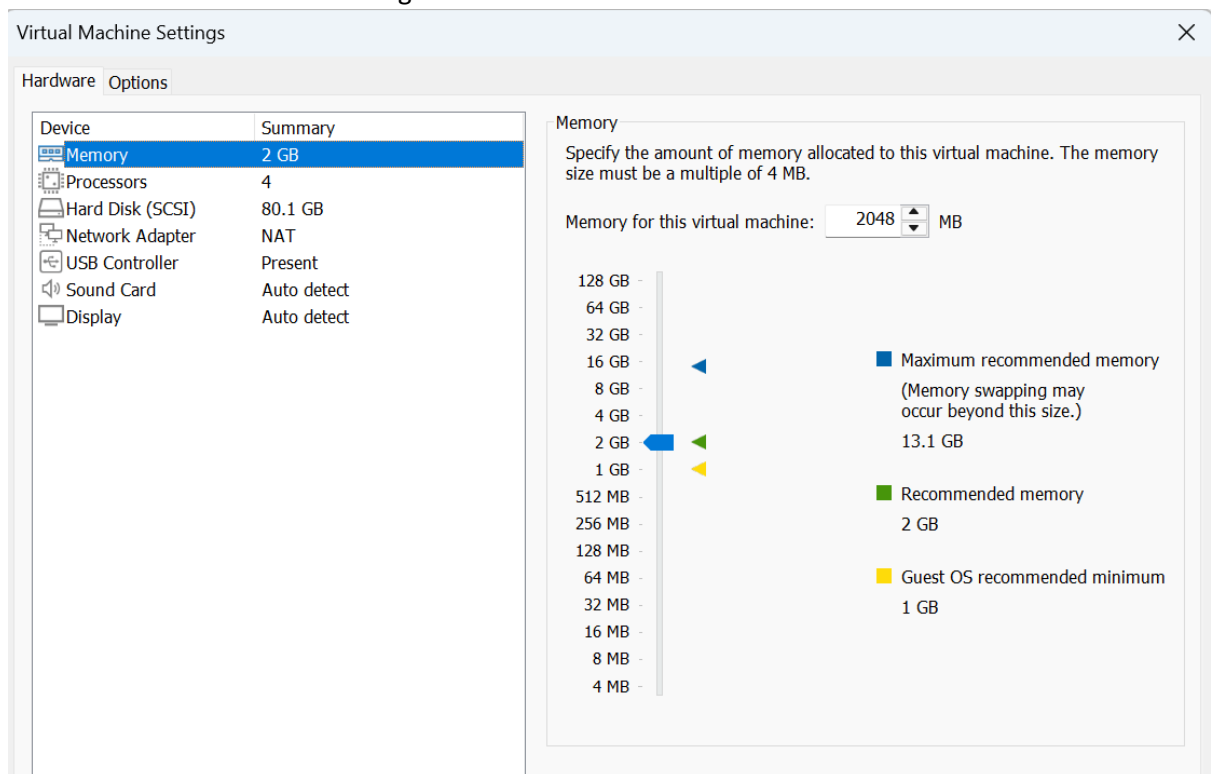


DAY 4 – SETTING UP KALI LINUX HOMELAB

AIM – Set up Kali Linux and DVWA, and install Nmap, Wireshark, BurpSuite, Gobuster, John, Hashcat and Autopsy.

STEPS :

1. Installing VMWare to set up Kali Virtual Machine – Installed VMWare Workstation Pro
2. Go to <https://www.kali.org/get-kali>
3. Choose Kali Platform as Virtual Machines
4. Choose VMWare Workstation as the Virtual Machine and install
5. This download needs to be unzipped – 7zip required! Download 7zip for Windows
6. Once Kali image is downloaded, open 7zip, locate the Kali Linux folder. Drag and drop the folder to where you wish to extract its contents. I have saved in a folder called Kali.
7. Open VMWare Workstation Pro
8. Click on “Open A Virtual Machine”
9. From Kali folder, select kali-linux-2026.1-vmware-amd64. VM gets opened.
10. Go to Edit Virtual Machine settings



ENSURE NAT IS USED TO SHARE THE HOST'S IP ADDRESS. Edited settings as shown above.

11. Turn on the VM
12. When turned on, enter initial username : kali, initial password : kali

13. Upon opening Kali Linux VM, we can see the pre-installed tools



14. Nmap, WireShark, BurpSuite, Hashcat, John, Hydra, Gobuster, and Autopsy, Metasploit-framework are already preinstalled in the VM.

15. Additionally, installing DVWA:

DVWA (Damn Vulnerable Web Application) is an intentionally vulnerable web application designed for practicing web-security concepts safely.

It gives you a legal target for learning things like:

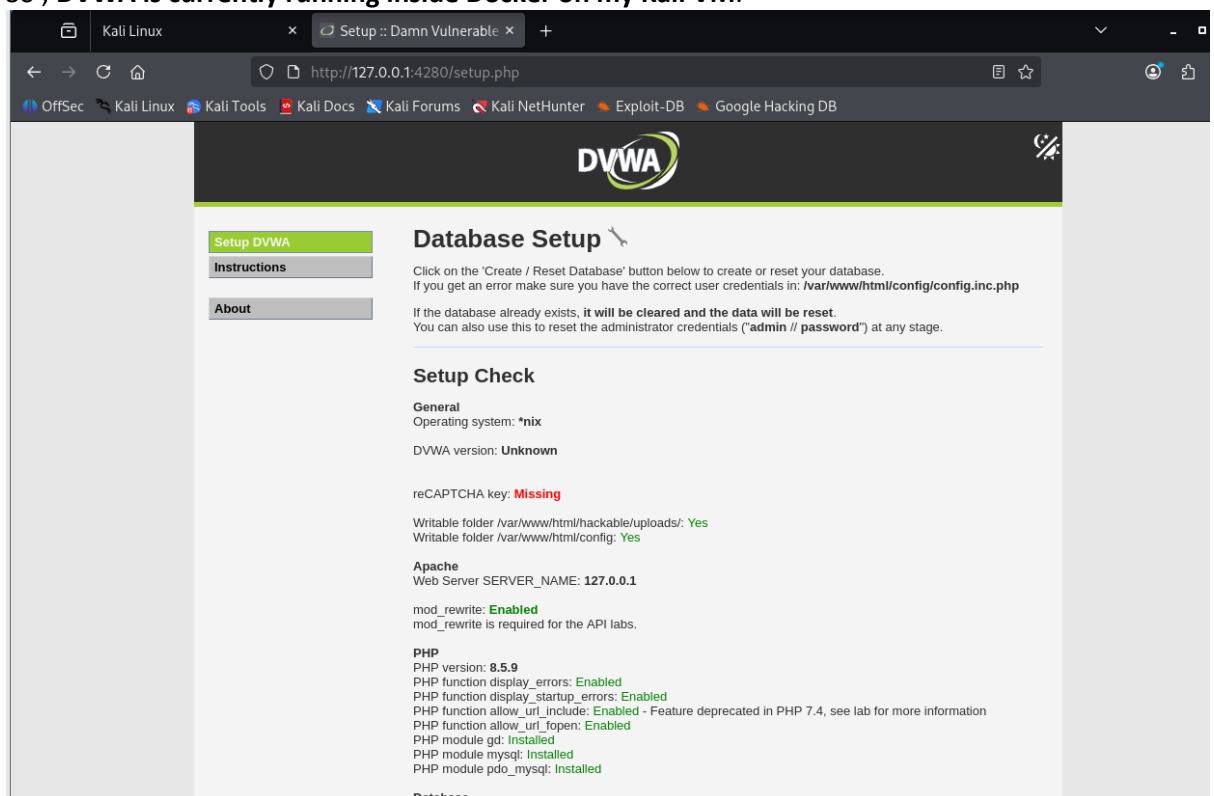
- SQL Injection
- XSS
- File Upload vulnerabilities
- Authentication weaknesses
- Command Injection
- Brute-force attacks
- CSRF
- Burp Suite testing

Your **Kali Linux VM** is **your security workstation**, but you need a deliberately vulnerable target to practice against. Instead of scanning or attacking real website, I can practice with DVWA.

DVWA setup:

1. `sudo apt update` – update kali

2. `sudo apt install docker.io -y` – install docker (Docker is used to run applications in isolated containers without having to install and configure all their dependencies directly on your system.)
3. `sudo systemctl enable --now docker` : start docker
4. `git clone https://github.com/digininja/DVWA.git` : clone DVWA
5. `cd ~/DVWA`
6. `sudo apt install docker-compose -y` : Docker compose for installing
7. `sudo docker-compose up -d` : start DVWA (-d means containers run in the background)
8. `sudo docker ps` : check that the container is running on localhost
9. Open in kali browser : <http://127.0.0.1:4280>
10. Username: admin Password: password
11. So , **DVWA is currently running inside Docker on my Kali VM.**



Kali is my **attacker/forensics workstation**. Tools like Nmap, Gobuster and Burp become much more useful when you have something you are actually authorized to scan and test.

DVWA: focused on web hacking → SQLi, XSS, authentication, file upload, etc.

Future work : Installing Metasploitable

Target	What you'll practice
DVWA	Web hacking, Burp, SQLi, XSS, authentication
Metasploitable	Nmap, enumeration, vulnerabilities, Metasploit
Kali	Your security/forensics workstation
Wireshark	Observe/analyze the traffic

CONCLUSION:

Successfully set up a Kali Linux cybersecurity homelab with essential security and digital forensics tools, along with DVWA running through Docker as an intentionally vulnerable web target. This setup provides a safe, isolated environment to practice network scanning, web vulnerability testing, traffic analysis, password attacks, and digital forensics without interacting with unauthorized systems. The lab will serve as the foundation for hands-on exercises.